

CHAPTER 10 Graphs

10.1 Graphs and Graph Models

10.2 Graph Terminology and Special Types of Graphs

10.3 Representing Graphs and Graph Isomorphism

10.4 Connectivity

10.5 Euler and Hamilton Paths

10.6 Shortest Path Problems

10.7 Planar Graphs

10.8 Graph Coloring



10.7 Planar Graphs

【Definition】 A graph is called *planar* if it can be drawn in the plane without any edges crossing .

Such a drawing is called a *planar representation* of the graph.

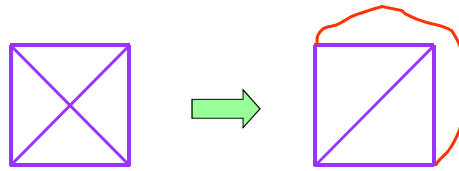
Understanding planar graph is important:

- Any graph representation of maps is planar.
- Electronic circuits usually represented by planar graphs.

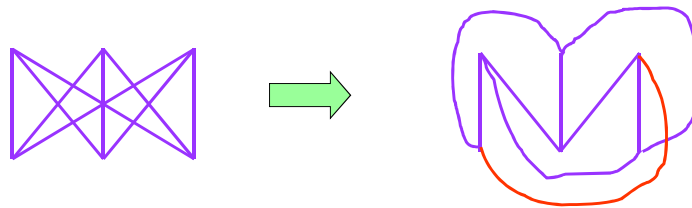


[[Example 1]] Are the following graphs planar?

(1)



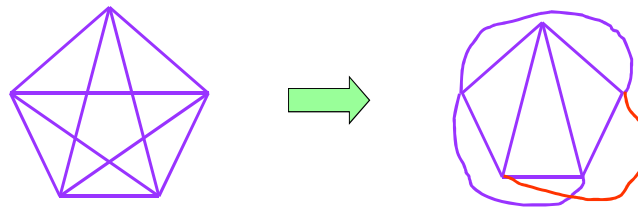
(2)



3



(3)



Note:

To prove that a graph is planar amounts to redrawing the edges in a way that no edges will cross. May need to move vertices around and the edges may have to be drawn in a very indirect fashion.

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1. Euler's Formula

Region

-- A region is a part of the plane completely disconnected off from other parts of the plane by the edges of the graph.

{ Bounded region
Unbounded region

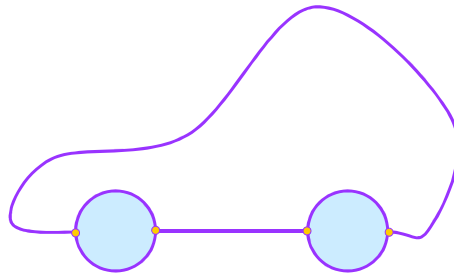
Note:

There is one unbounded region in a planar graph.



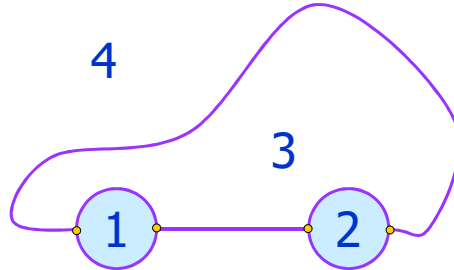
[[Example 2]] How many regions does the following graph have?

(1)

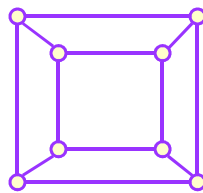


[[Example 2]] How many regions does the following graph have?

(1)

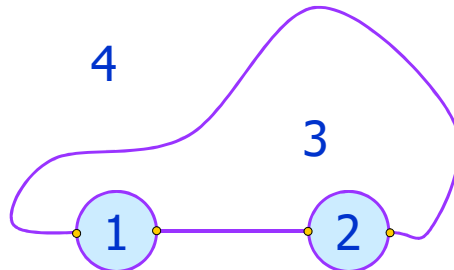


(2)

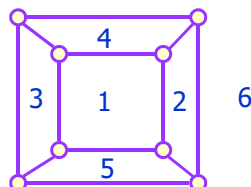


[[Example 2]] How many regions does the following graph have?

(1)



(2)



【 Theorem 1 】 Euler's formula

Let G be a **connected planar simple** graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Then $r=e-v+2$.

Proof:

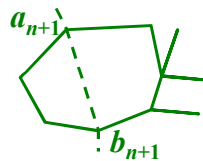
First, we specify a planar representation of G . We will prove the theorem by constructing a sequence of subgraphs $G_1, G_2, \dots, G_e = G$, successively adding an edge at each stage.

The constructing method: Arbitrarily pick one edge of G to obtain G_1 . Obtain G_n from G_{n-1} by arbitrarily adding an edge that is incident with a vertex already in G_{n-1} , adding the other vertex incident with this edge if it is not already in G_{n-1} .

Let r_n, e_n , and v_n represent the number of regions, edges, and vertices of the planar representation of G_n induced by the planar representation of G , respectively.



- (1) The relationship $r_1 = e_1 - v_1 + 2$ is true for G_1 , since $e_1=1, v_1=2$, and $r_1=1$.
 - (2) Now assume that $r_n = e_n - v_n + 2$. Let $\{a_{n+1}, b_{n+1}\}$ be the edge that is added to G_n to obtain G_{n+1} .
- ◆ Both a_{n+1} and b_{n+1} are already in G_n .



These two vertices must be on the boundary of a common region R , or else it would be impossible to add the edge $\{a_{n+1}, b_{n+1}\}$ to G_n without two edges crossing (and G_{n+1} is planar).

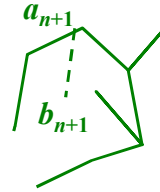
The addition of this new edge splits R into two regions.

Consequently, $r_{n+1} = r_n + 1$, $e_{n+1} = e_n + 1$, and $v_{n+1} = v_n$. Thus, $r_{n+1} = e_{n+1} - v_{n+1} + 2$.



- ◆ One of the two vertices of the new edge is not already in G_n .

Suppose that a_{n+1} is in G_n but that b_{n+1} is not.



Adding this new edge does not produce any new regions, since b_{n+1} must be in a region that has a_{n+1} on its boundary.

Consequently, $r_{n+1} = r_n$. Moreover, $e_{n+1} = e_n + 1$ and $v_{n+1} = v_n + 1$.

Hence, $r_{n+1} = e_{n+1} + 1 - v_{n+1} - 1 + 2$.

Note:

- 1) The Euler's formula is a *necessary condition*.
- 2) How about unconnected simple planar graph?



【Example 3】 Suppose that a connected planar simple graph has 20 vertices, each of degree 3. How many regions does this planar graph have?

Solution:

By handshaking theorem,

$$3v = 2e \Rightarrow e = 30$$

From Euler's formula, the number of regions is

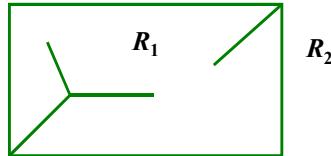
$$r = e - v + 2 = 30 - 20 + 2 = 12$$



【Definition】 Suppose R is a region of a connected planar simple graph, **the number** of the edges on the boundary of R is called the **Degree of R** .

Notation: $\text{Deg}(R)$

For example,



$$\text{Deg}(R_1) = 12$$

$$\text{Deg}(R_2) = 4$$



【Corollary 1】 If G is a connected planar simple graph with e edges and v vertices where $v \geq 3$, then $e \leq 3v - 6$

Proof:

Suppose that a connected planar simple graph divides the plane into r regions, the degree of each region is at least 3.

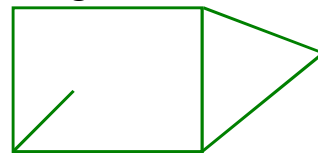
$$\text{Since } 2e = \sum \text{deg}(R_i) \geq 3r,$$

it imply $r \leq (2/3)e$

Using Euler's formula $e - v + 2 = r$, we obtain

$e - v + 2 \leq (2/3)e$, this shows that

$$e \leq 3v - 6$$



Note:

- ◆ The equality holds if and only if every region has exactly three edges.
- ◆ For unconnected planar simple graph, $e \leq 3v - 6$ also holds.

Since for a component, $e_i \leq 3v_i - 6$

$$e = \sum e_i \leq \sum (3v_i - 6) < 3 \sum v_i - 6 = 3v - 6$$



【 Corollary 2 】 If G is a connected planar simple graph, then G has a vertex of degree not exceeding five.

Proof:

- (1) G has one or two vertices
- (2) G has at least three vertices

By Corollary 1, we know that $e \leq 3v - 6$, so $2e \leq 6v - 12$

If the degree of every vertex were at least six, then

$$2e \geq 6v$$



【 Corollary 3 】 If a connected planar simple graph has e edges and v vertices with $v \geq 3$ and no circuits of length 3, then $e \leq 2v - 4$.

Generally, if every region of a connected planar simple graph has at least k edges, then

$$e \leq \frac{(v-2)k}{k-2}$$

$$r = e - v + 2$$

$$kr \leq 2e$$

$$r \leq 2e/k$$

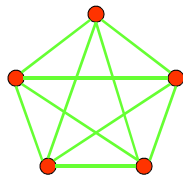
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【 Example 4 】 Show that K_5 , $K_{3,3}$ are nonplanar.

Proof:

(1)



The graph K_5 has 5 vertices and 10 edges.

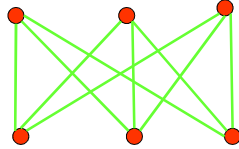
However, the inequality $e \leq 3v - 6$ is not satisfied for this graph since $e=10$ and $3v-6=9$.

Therefore, K_5 is not planar.

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(2)



$K_{3,3}$ has 6 vertices and 9 edges.

Since $K_{3,3}$ has no circuits of length 3 (this is easy to see since it is bipartite), Corollary 3 can be used .

Since $e=9$ and $2v-4=8$, corollary 3 shows that $K_{3,3}$ is nonplanar.

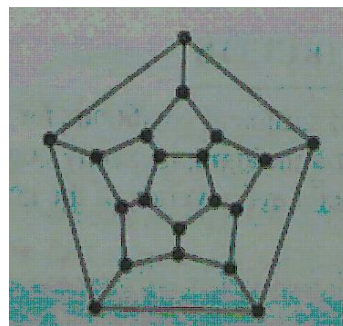


[[Example 5]] The construction of Dodecahedron .

Solution:

Since the degree of every vertex is 3 and the degree of every region is 5. Then

$$\left\{ \begin{array}{l} 2e = 3v \\ 2e = 5r \\ r = e - v + 2 \end{array} \right.$$

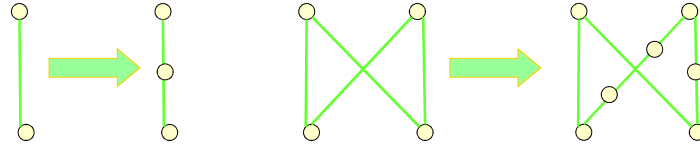


It follows that $v=20$, $e=30$ and $r=12$.



2. KURATOWSKI'S THEOREM

Elementary subdivision

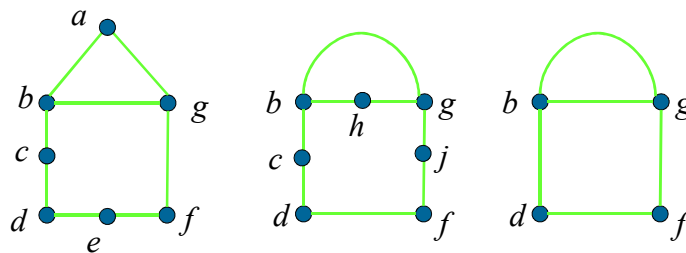


Homeomorphic

-- The graph $G_1=(V_1,E_1)$ and $G_2=(V_2,E_2)$ are called homeomorphic if they can be obtained from the same graph by a sequence of elementary subdivision.



For example,

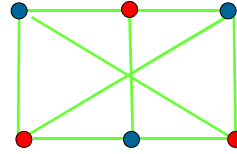
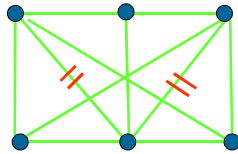


【 Theorem 2 】 A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5 .

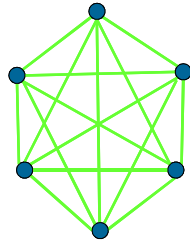


[[Example 6]] Determine whether the following graphs are planar.

(1)

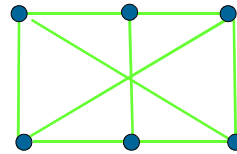
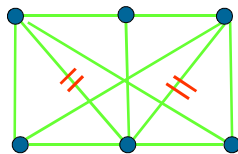


(2)

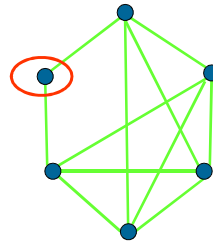
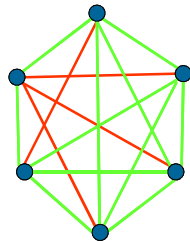


[[Example 6]] Determine whether the following graphs are planar.

(1)

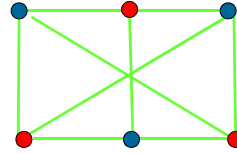
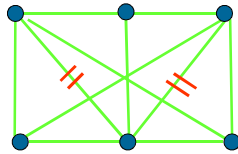


(2)

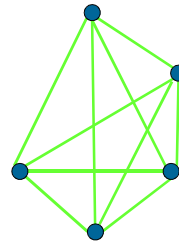
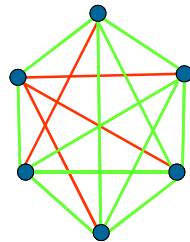


[[Example 6]] Determine whether the following graphs are planar.

(1)

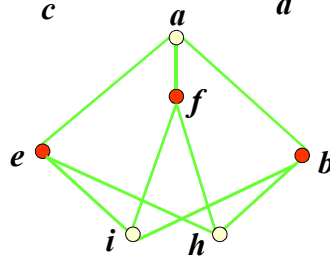
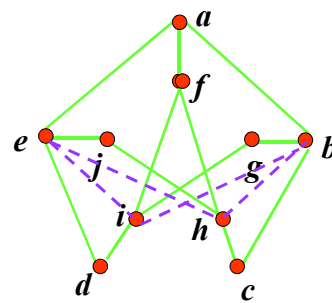
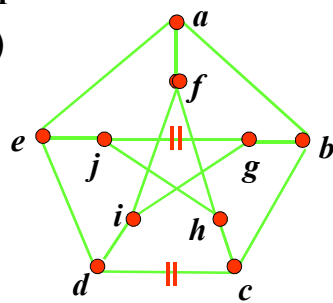


(2)



[[Example 6]] Determine whether the following graphs are planar.

(3)



Homework:

Sec. 10.7 7, 20, 22, 23, 25



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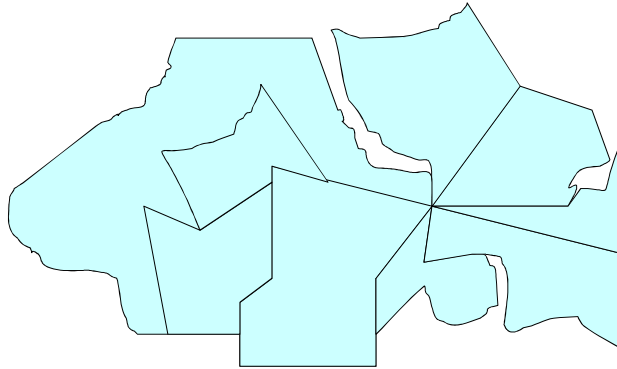
10.7 Planar Graphs

10.8 Graph Coloring



The problem related to the coloring of maps:

Determining the least number of colors that can be used to color a map so that adjacent regions never have the same color.

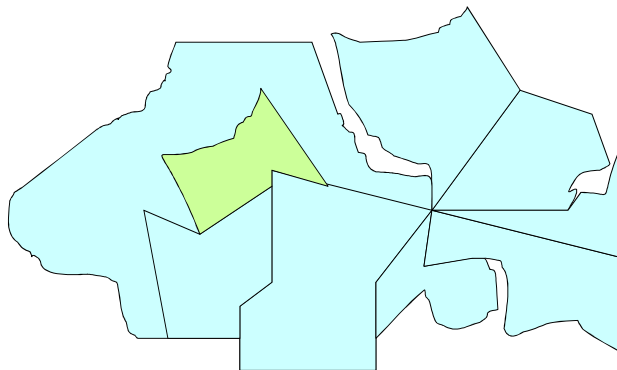


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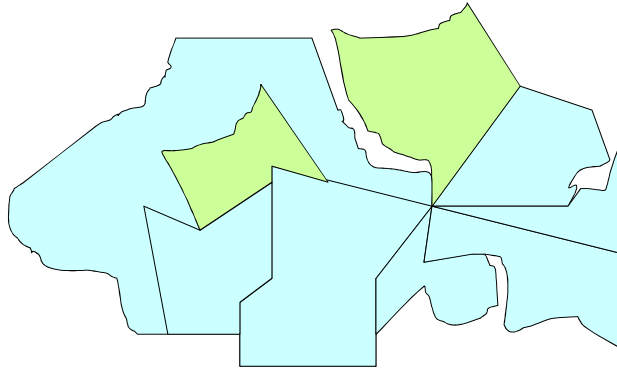


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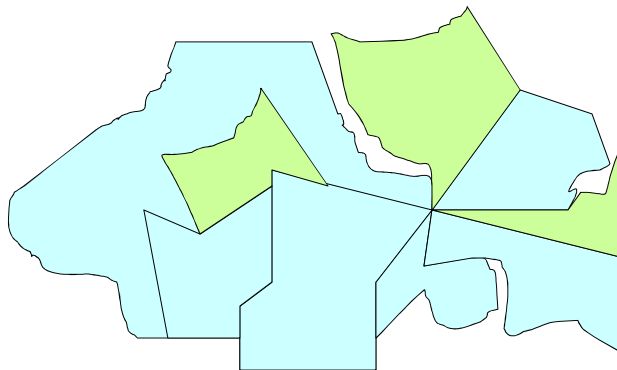


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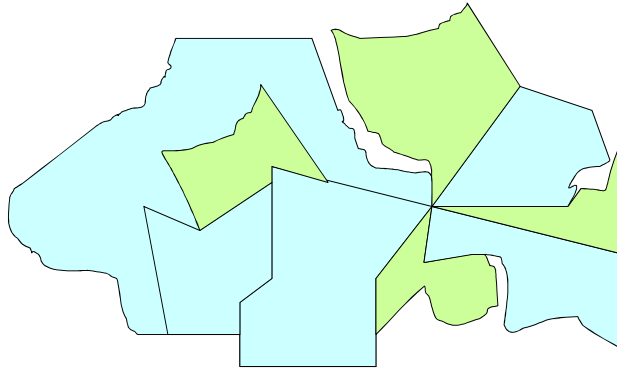


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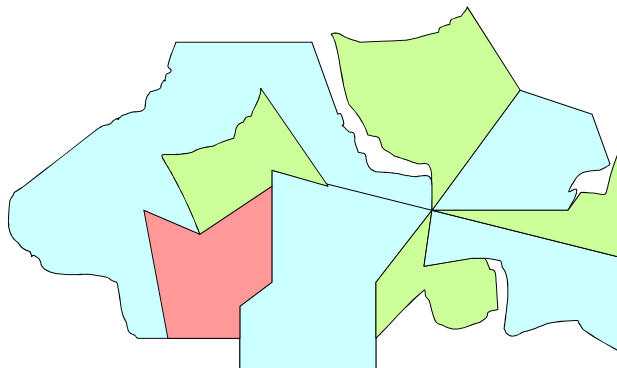


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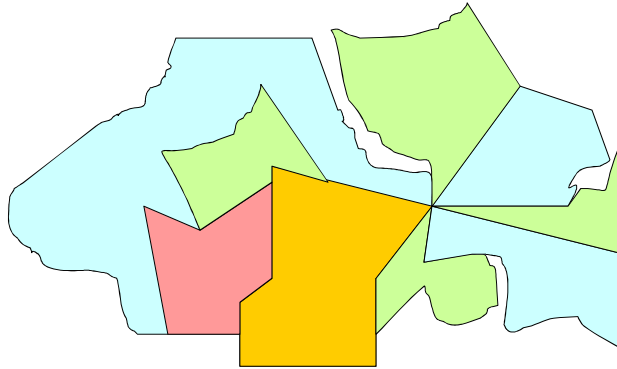


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The problem related to the coloring of maps:

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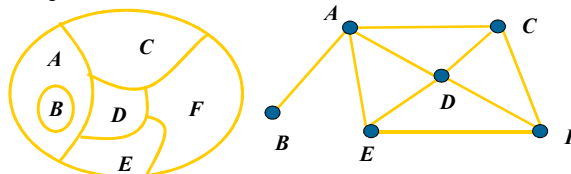


The problem of coloring a map, can be reduced to a graph-theoretic problem.

Each map in the plane can be represented by a graph, namely *the dual graph of the map*.

- Each region of the map is represented by a vertex.
- An edge connect two vertices if the regions represented by these vertices have a common border.
- Two regions that touch at only one point are not considered adjacent.

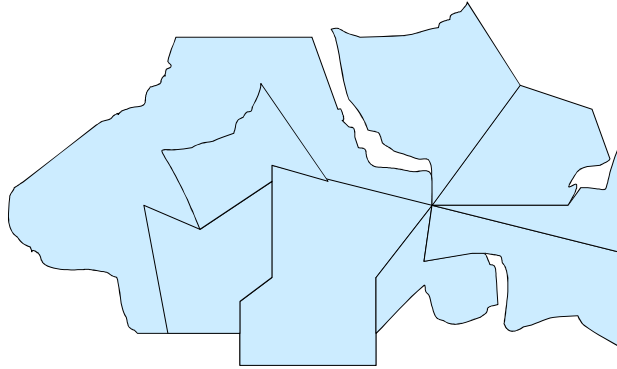
For example,



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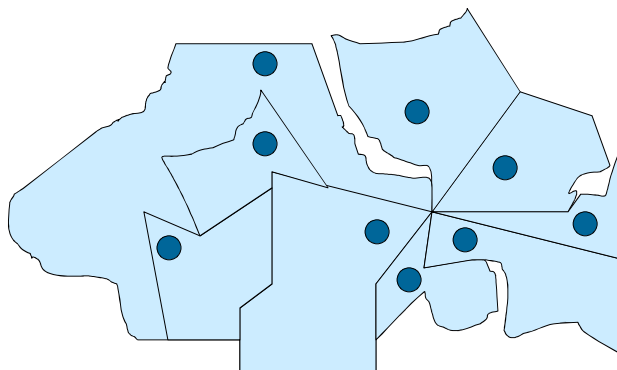
Coloring a map is equivalent to coloring the vertices of its dual graph so that no two adjacent vertices in this graph have the same color.



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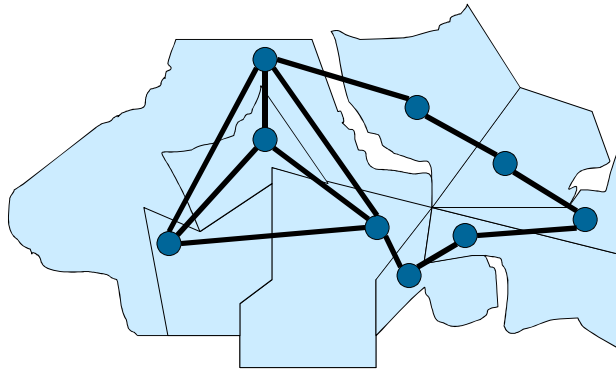
For each region introduce a vertex.



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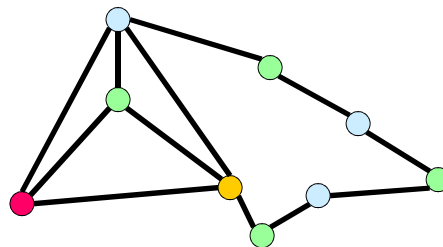
For each pair of regions with a positive-length common border introduce an edge.



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Coloring regions of a map is equivalent to coloring vertices of its dual graph.



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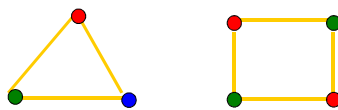
*Terminologies:**coloring*

- A *coloring* of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

The chromatic number of a graph

- is the least number of colors needed for a coloring of this graph, denoted by $\chi(G)$

For example,

**【 Theorem 1 】 The Four Color Theorem**

The chromatic number of a **planar graph** is no greater than four.

- ◆ Any planar map of regions can be depicted using 4 colors so that no two regions with a common border have the same color.
- ◆ The four color theorem was originally proposed as a conjecture in 1850s.
- ◆ Proof by Haken and Appel used exhaustive computer search in 1976.
- ◆ The four color theorem applies only to planar graphs. Nonplanar graphs can have arbitrarily large chromatic numbers.



1. The chromatic numbers of some simple graphs

Show that the chromatic numbers of a graph is n .

- Show that the graph can be colored with n colors.

Method: constructing such a coloring.

- Show that the graph cannot be colored using fewer than n colors.

The chromatic numbers of some simple graphs:

- (1) The graph G contains only some isolated vertices.

$$x(G) = 1$$

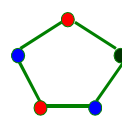
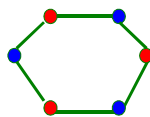


- (2) The graph G is a path containing no circuit.

$$x(G) = 2$$



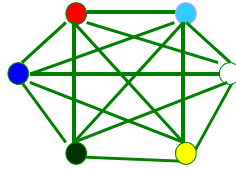
- (3) C_n



$$\left\{ \begin{array}{ll} x(G) = 2 & \text{if } n \text{ is even} \\ x(G) = 3 & \text{if } n \text{ is odd} \end{array} \right.$$



(4) K_n



$$\chi(G) = n$$

(5) A simple graph with a chromatic number of 2 is bipartite.
A connected bipartite graph has a chromatic number of 2.



Homework:

Sec. 10.8 3, 8, 9, 10

